AoI Energy-Efficient Edge Caching in AAV-Assisted Vehicular Networks

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***Abstract*—Mobile edge caching (MEC) has grown substantially with the rapid development in scale and complexity of data traffic. By exploiting the expansive coverage of autonomous aerial vehicles (AAVs), MEC enables services for massive vehicle users (VUs) simultaneously, which is promising for enhancing network transmission efficiency. Nonetheless, due to challenges arising from the timeliness and freshness of content services caused by AAVs' limited endurance and airborne capacity, caching strategy considering the real-time of content in large-scale dynamic Internet of Vehicles (IoV) environments remains open. With the above consideration, in this article, the cache refreshing cycle and content placement are jointly optimized in the cache-enabled AAV-assisted vehicular integrated networks (CAVINs) to minimize the content Age of Information (AoI) and energy consumption of the macro AAV. Since the joint optimization problem is variational coupled with nonconvex binary constraints, it is decoupled and solved by a double-iteration method. Specifically, the optimal cache refreshing cycle is derived in semi-closed form with the Karush-Kuhn-Tucker (KKT) conditions. The locally optimal solution of the content placement is obtained through successive convex approximation (SCA). Simulation results corroborate the effectiveness and superiority of the proposed scheme.**

***Index Terms*—Age of Information (AoI), Internet of Vehicles (IoV), mobile edge caching (MEC), unmanned aerial vehicles (AAVs)-assisted networks.**

I. Introduction

THE RAPID development of vehicle-to-everything (V2X) [1] and wireless mobile communication technologies [2], [3] has significantly flourished various intelligent services and applications (e.g., intelligent road environment perception, vehicle behavior control, image-aided navigation, human-vehicle interactions, etc.) [4], [5] to greatly improve the experience for vehicle users (VUs). However, due to multiple users subscribing to these services simultaneously [6], content requirements for ubiquitous network coverage, massive storage space, and robust security guarantee to supply low-latency, high-scalability, and high-reliability demands are pushing the limits of current wireless systems [7]. Fortunately, riding on the wave of technological development of the Internet of Vehicles (IoV), mobile edge caching (MEC) is introduced as a promising paradigm that can better support time-critical vehicular applications [8]. Specifically, MEC enables content services in proximity to the storage resources of radio access facilities and mobile devices, bringing tremendous benefits to users and operators [9], [10], [11].

Despite the attractive benefits, MEC-enabled vehicular networks rely heavily on network infrastructures. Therefore, it is a challenge for content caching in the areas with insufficient network facilities [12]. This ignites researchers to leverage flexible and cost-effective wireless connectivity of unmanned aerial vehicles (AAVs) to meet the requirements of highly dynamic vehicular networks [13]. Due to their controllable mobility and altitude, AAVs can support rapid deployment and stable communication with a high probability of Line of Sight (LoS) for AAV-to-vehicle (A2V) links [14], [15].

Inspired by the introduction of A2V, AAV-assisted content caching is envisioned as a key booster to serve terrestrial vehicles flexibly. In view of its appealing features, substantial research efforts have been devoted to AAV-assisted content caching. Ji et al. [16] focused on the performance analysis and trajectory optimization of cache-enabled AAV-assisted device-to-device networks to maximize the cache hit probability. A framework of cache-enabled AAV nonorthogonal multiple access (NOMA) cellular networks for content delivery to ground users in a hot spot area was proposed in [17]. Besides, to maximize the throughput and utilization of AAVs with low-power consumption, a structure of cooperative Internet of Things (IoT) networks assisted by cache-enabled AAVs was investigated in [18]. Chai and Lau [19] proposed an online AAV-assisted wireless caching design via joint optimization AAV trajectory, transmission power, and content caching scheduling. In addition, as AAV-assisted content caching has brought remarkable changes to MEC, the energy minimization problem of AAVs with limited endurance and onboard energy has attracted much attention. For example, a closed-form propulsion power consumption model aiming at minimizing the total energy consumption through joint optimization of trajectory and time scheduling among ground users for AAVs was derived in [20]. Based on the energy model in [20] and [21]

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investigated the AAV trajectory to minimize the total energy consumption while satisfying the time-out requirement and energy budget.

Nonetheless, MEC may lead to staleness in dynamic items whose content information changes with time and environment [22]. On one hand, caching content with high dynamics introduces frequent cache refreshing, which degrades the system transmission efficiency and results in high-AAV energy consumption [23]. On the other hand, outdated updates may be inconsistent with the current status of the monitored and controlled process, leading to erroneous decisions [24]. In this sense, Hu et al. [25] proposed the concept of Age of Information (AoI) to quantify the timeliness of the obtained status updates. A higher value of AoI indicates that the status update is of less value for the accuracy of the environmental perception. Although there have already been some recent pioneering efforts considering AoI in A2V networks, few of them investigated the tradeoff between AAVs' energy consumption and the content refreshing problems in MEC. For example, taking the limited endurance of AAVs into account, the reinforcement learning-based trajectory optimization method was proposed in [26] to minimize the AoI subject to energy. AoI was utilized as a performance metric to quantify the temporal correlation between data packets in [27], as well as an AoI-energy-aware data collection scheme for AAV-assisted IoT networks was explored. Nevertheless, these pioneering works either only introduced AoI as a performance metric without digging into the relationships between the content placement and refreshing or just concentrated on the static content in AAV-assisted MEC neglecting the dynamic real-time requirements for content. The growing demand for real-time content from users would present a significant challenge for lightweight caching AAVs in situations of emergent traffic congestion scenes on suburban or rural roads, particularly in the presence of damaged or under-construction roadside units or base stations. Additionally, these AAVs may be overwhelmed by caching a large amount of content that exceeds their limited capacity. It is therefore of significant importance to introduce a macro AAV to alleviate the burden of lightweight AAVs, while also maintaining real-time content. Concurrently, the energy consumption of the macro AAV may increase for refreshing and delivering real-time content. Due to these difficulties, to the best of our knowledge, the AAV-assisted MEC problem of balancing AAV energy consumption and content AoI in the IoV remains open.

Therefore, to improve the efficiency and quality of AAV-assisted content caching, it is necessary to jointly consider the AAV energy consumption and content freshness in large-scale dynamic IoV environments. With the above consideration, targeting remote suburban scenarios with non-substantial infrastructures, a joint optimization problem in the cache-enabled AAV-assisted vehicular integrated networks (CUVINs) is studied in this work. The main contributions of this article are summarized as follows.

1) An optimization problem in the CUVIN is innovatively formulated to minimize the weighted sum of the content AoI and energy consumption of the macro AAV, which jointly considers the cache refreshing cycle and cache content placement.

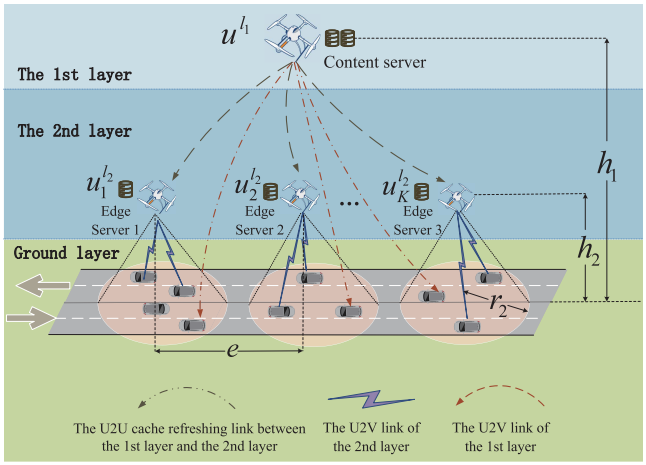


Fig. 1. Scenario of cache-enabled AAV-assisted vehicular networks.

2) An alternating optimization caching strategy (AOCS) is proposed to efficiently solve the mixed integer nonlinear programming (MINP) minimization problem jointly considering the impact of content popularity, content size, request frequency, traffic flow, the total number of content, and edge layer AAV's cache capacity.
3) It is verified that the proposed scheme decreases the expected AoI and energy consumption compared with other schemes. The necessity of jointly optimizing the cache refreshing cycle and content placement to enhance the performance of the CUVIN is also demonstrated.

The remainder of this article is organized as follows. In Sections II and III, the system model and the performance of AoI are introduced, respectively. Followed by the problem analysis and the proposed algorithms in Section IV. The simulations are shown in Section V. Finally, the conclusion and future works are presented in Section VI.

II. System Model

An edge caching system in a suburban area without sufficient infrastructure is considered. As shown in Fig. 1, a 1-D road consisting of J two-way lanes is considered where VUs moving at a constant speed are uniformly distributed. The density of VUs on each lane is denoted by ρ_v, which is assumed to be quasi-static during the scheduling period. Lightweight AAVs hover above the roads and serve VUs within their coverage with content demands, such as the status of traffic jams, the availability of a petrol station, and the real-time 3-D map of the environment. To meet the increasing real-time demands of the VUs, the macro AAV is introduced with the ability to produce and refresh content, hovering above the entire area and providing support [18]. Assuming that the AAVs hover over the VUs until edge caching ends, the energy consumption optimization of the macro AAV for hovering and operation is omitted.

The AAVs at the second layer with a height h_2, represented by a set $\mathcal{U}^{l_2} = \{u_1^{l_2}, u_2^{l_2}, \ldots, u_k^{l_2}, \ldots, u_K^{l_2}\}$, are uniformly distributed over the road with distance e, acting as edge servers for lightweight caching. The cache-enabled AAVs of $\mathcal{U}^{l_2}$ deliver VUs' requested content (such as driving-related content or real-time ambient content) if they are cached. Otherwise,

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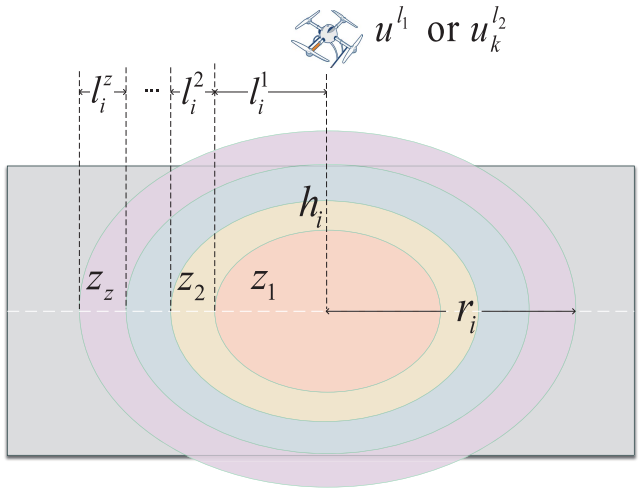


Fig. 2. Zone-based transmission model.

they will be served by the macro AAV server u^{l_1}. The macro AAV server u^{l_1} at the first layer with a height h_1 ensures the stable coverage for all VUs within the whole range area and provides cache delivery services for VUs outside the service of the second layer's AAVs as well as periodically refreshes the content cached at the second layer's AAVs.

A. A2V Transmission Model

It is assumed that the A2V links adopt the Wi-Fi spectrum to communicate [28]. In the CUVIN, AAVs are equipped with directional antennae with uniform beam width θ to serve VUs within the circular coverage area, whose radii satisfy

$$r_i \le h_i \tan\frac{\theta}{2} \tag{1}$$

where $i \in \{1, 2\}$ is the layer of AAVs. Furthermore, the coverage area is divided into Z zones, as shown in Fig. 2, represented by $\mathcal{Z} = \{z_1, z_2, \ldots, z_z, \ldots, z_Z\}$, each zone z_z has a distinct annulus area with width l_i^z. Wi-Fi channels are shared by associated VUs through a competitive mechanism.

The communication between AAVs and VUs within the coverage area is considered air-to-ground communication, which consists of two transmission channels [16]: 1) LoS and 2) Non-LoS (NLoS) channels. For A2V links, the LoS probability is given by

$$\mathbb{P}_{\text{LoS}}\left(d_i^z, h_i\right) = \frac{1}{1 + a_1 \exp\left(-a_2\left(\arctan\left(\frac{h_i}{d_i^z}\right) - a_1\right)\right)} \tag{2}$$

where a_1 and a_2 are constants determined by the environment [29]. d_i^z is the horizontal distance between the AAV at layer i and the VU in the zone z_z.

The average path loss of A2V links can be expressed as follows [30]:

$$\begin{aligned}\text{PL}\left(d_i^z, h_i\right) = {} & \mathbb{P}_{\text{LoS}}\left(d_i^z, h_i\right) \cdot \left(\eta_{\text{LoS}} - \eta_{\text{NLoS}}\right) \\ & + 10\log\left(\left(d_i^z\right)^2 + h_i^2\right) + 20\log\left(\frac{4\pi f_c}{c}\right) + \eta_{\text{NLoS}}\end{aligned} \tag{3}$$

where c, f_c, η_{NLos}, and η_{Los}, respectively, represents the speed of light, the carrier frequency, the shadowing, and diffraction losses for LoS and NLoS links, respectively.

Thus, the average rate $\mathcal{R}_i^z$ between a VU in the region z_z and a AAV can be given by

$$\mathcal{R}_i^z = \mathop{\mathbb{E}}_{d_i^z, h_i}\left[B_i \log\left(1 + \frac{P_i}{\sigma^2 \cdot 10^{\text{PL}(d_i^z, h_i)/10}}\right)\right]. \tag{4}$$

P_i is the transmit power of the AAV in layer i, σ^2 is the noise power, and B_i represents the bandwidth of the A2V link of layer i. To make full use of AAVs, the AAVs of $\mathcal{U}^{l_2}$ do not overlap and cannot communicate with each other, therefore the interference is negligible.

Therefore, the average data rate in the AAV coverage area is

$$\mathcal{R}_i = \varrho\left\{\frac{\sum_{z=1}^{Z}\left(\mathcal{R}_i^z\left[\left(l_i^z + l_i^{z-1}\right)^2 - \left(l_i^{z-1}\right)^2\right]\right)}{\left(\sum_{z=1}^{Z} l_i^z\right)^2}\right\} \tag{5}$$

where ϱ is the throughput efficiency factor [29].

B. Content Request and Caching Model

It is assumed that content size follows the Poisson distribution with mean value s. That is, for the content $\mathcal{F} = \{f_1, f_2, \ldots, f_f, \ldots, f_F\}$, with the corresponding size $\mathcal{S} = \{s_1, s_2, \ldots, s_f, \ldots, s_F\}$, where F represents the total amount of generated content and s_f represents the size of content f_f. Moreover, the content popularity denoted by $\mathcal{P} = \{p_1, p_2, \ldots, p_f, \ldots, p_F\}$, follows the Zipf distribution [31] as:

$$p_f = \frac{f^{-\mu}}{\sum_{f=1}^{F} f^{-\mu}} \tag{6}$$

where μ is the Zipf distribution parameter.

A continuous-time model is utilized to analyze the service process of content requests and delivery. The content request follows a Poisson process. The arrival rate of each VU's request is represented by λ. Thus, the aggregate request arrival processes of the VUs at u^{l_1} and the AAVs of $\mathcal{U}^{l_2}$ also follow Poisson processes with rates Λ_i, $i \in \{1, 2\}$, which can be given by [32]:

$$\Lambda_i = N_i\lambda = 2J \cdot r_i\rho_v\lambda \tag{7}$$

where N_i, $i \in \{1, 2\}$ is the number of VUs in the coverage area of AAVs at layer i. Then the requests of the VUs for content f_f follow the Poisson process with strength Λ_i^f given by:

$$\Lambda_i^f = p_f \cdot \Lambda_i. \tag{8}$$

Assuming AAVs of $\mathcal{U}^{l_2}$ cache the same content. Denote $\mathcal{C} = \{c_1, c_2, \ldots, c_f, \ldots, c_F\}$ as the content placement collection, where $c_f \in \{0, 1\} \quad \forall f$. Specifically, $c_f = 1$ means that all AAVs of $\mathcal{U}^{l_2}$ have cached the content f_f, so that a VU can fetch the content f_f directly from its associated AAV $u_k^{l_2} \in \mathcal{U}^{l_2}$. Otherwise, $c_f = 0$ means all AAVs of $\mathcal{U}^{l_2}$ have not cached the content f_f, so the content requested by the VU will be delivered through the AAV u^{l_1}. Each AAV at the edge layer has the

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same limited cache capacity $S_c^{l_2}$, and the capacity constraint is given by

$$\sum_{f=1}^{F} s_f c_f \le S_c^{l_2}. \tag{9}$$

C. Model of Energy Consumption of the Macro AAV

Record the collection of cache refreshing cycles as $\Delta_{\text{ref}} = \{\Delta_1, \Delta_2, \ldots, \Delta_f, \ldots, \Delta_F\}$, where Δ_f represents the cache refreshing cycle of f_f. When the content has cached in the AAVs of $\mathcal{U}^{l_2}$, the macro AAV is triggered to send new versions to the AAVs of $\mathcal{U}^{l_2}$ periodically with refreshing cycle Δ_f.

The average energy consumption Φ of the macro AAV u^{l_1} is expressed as

$$\Phi = \frac{1}{F}\sum_{f=1}^{F} \Phi_f^{l_1}\left(1 - c_f\right) + \frac{1}{F}\sum_{f=1}^{F} \Phi_f^{l_2} c_f \tag{10}$$

where $\Phi_f^{l_1} = \Lambda_1^f s_f \varepsilon_f$ is the energy consumption consumed by u^{l_1} to deliver content f_f to VUs when AAVs of $\mathcal{U}^{l_2}$ have not cached the content ($c_f = 0$) and cannot serve VUs, where ε_f represents the energy consumption consumed by u^{l_1} to transmit per unit of content. $\Phi_f^{l_2} = (s_f \varepsilon_f / \Delta_f)$ denotes the energy consumption consumed by u^{l_1} to update content f_f that has been cached in the AAVs of $\mathcal{U}^{l_2}$ ($c_f = 1$).

III. Performance of AoI

The AoI of content served by AAVs from two layers will be discussed, specifically, the AoI $\Delta_f^{l_2}$ of content f_f served by the AAV $u_k^{l_2} \in \mathcal{U}^{l_2}$ is considered first. Then the AoI $\Delta_f^{l_1}$ is investigated when the content f_f is not cached in the AAV $u_k^{l_2} \in \mathcal{U}^{l_2}$ and need to be delivered by u^{l_1}. The received content requests are served in a first-in–first-out (FIFO) manner. The content service of AAVs at layer i can be modeled as an M/M/1 queuing model [33] with arrival rates $\sum_{f=1}^{F} \Lambda_1^f$ and $\sum_{f=1}^{F} \Lambda_2^f \cdot c_f$, and service rate $\mathcal{R}_i/s$.

A. AoI of Content Served by the Second Layer

Considering the transfer of content between the AAV $u_k^{l_2}$ and the VUs within its coverage area, the aggregation requests of the VUs follow a Poisson process with the parameter Λ_2. In this case, the AoI $\Delta_f^{l_2}$ can be expressed as follows:

$$\Delta_f^{l_2} = \mathbb{E}\left[\Delta_f^d\right] + \mathbb{E}\left[\Delta_{2,w,f}\right] \tag{11}$$

where difference time Δ_f^d refers to the difference between the time when the last update of the content arrives at the AAV and the time when the VUs' requests reach the AAV, and $\Delta_{2,w,f}$ denotes the queue waiting time (service latency) of VUs for content f_f served by AAVs of the second layer $\mathcal{U}^{l_2}$.

To obtain the difference time Δ_f^d, the time slot $[0, \Delta_f]$ is utilized to describe the request's arrival time. N_2 VUs are served with the requested content f_f. t_n is the arrival time of the nth VU's requested content f_f during the cache refreshing cycle Δ_f. Since VUs' requests for content f_f follow the Poisson process of the strength Λ_2^f, t_n obeys a uniform distribution within $[0, \Delta_f]$.

Lemma 1: The average AoI $\Delta_f^{l_2}$ of content f_f is expressed as

$$\Delta_f^{l_2} = \frac{\Delta_f}{2} + \frac{1}{\frac{\mathcal{R}_2}{s} - \sum_{f=1}^{F} 2Jr_2\rho_v\lambda p_f c_f}. \tag{12}$$

Proof: Please refer to Appendix A. ■

B. AoI of Content Served by the First Layer

Assuming that all requests of VUs for content cached in u^{l_1} obey the Poisson process with strength Λ_1. If the content f_f is served by the macro AAV u^{l_1}, the average AoI $\Delta_f^{l_1}$ of content f_f can be given by similar steps as in Appendix A as follows:

$$\Delta_f^{l_1} = \frac{1}{\frac{\mathcal{R}_1}{s} - \sum_{f=1}^{F} \Lambda_1^f} = \frac{1}{\frac{\mathcal{R}_1}{s} - \sum_{f=1}^{F} 2Jr_1\rho_v\lambda p_f}. \tag{13}$$

Therefore, the AoI of content in the edge caching network can be given by

$$\Delta = \sum_{f=1}^{F} p_f\left(c_f\Delta_f^{l_2} + \left(1 - c_f\right)\Delta_f^{l_1}\right). \tag{14}$$

IV. Joint Optimization Issues for Cached Content Refreshing Cycles and Content Placement

The weighted sum Ω of the content AoI Δ and energy consumption Φ of the macro AAV u^{l_1} is optimized. Accordingly, the weighted sum Ω in the considered CUVIN is as follows:

$$\Omega = \alpha\sum_{f=1}^{F} p_f\left(c_f\Delta_f^{l_2} + \left(1 - c_f\right)\Delta_f^{l_1}\right) + (1 - \alpha)\cdot\Phi \tag{15}$$

where $\alpha > 0$ is the weighting factor.

Therefore, the joint optimization problem is formulated as

$$\mathbf{P1}: \min_{\mathcal{C}, \Delta_{\text{ref}}} \quad \Omega \tag{16a}$$

$$\text{s.t.} \quad \sum_{f=1}^{F} c_f/\Delta_f \le M \tag{16b}$$

$$\sum_{f=1}^{F} c_f s_f \le S_c^{l_2} \tag{16c}$$

$$\sum_{f=1}^{F} \Lambda_1^f c_f \le \frac{\mathcal{R}_2}{s} \tag{16d}$$

$$c_f \in \{0, 1\} \quad \forall f \tag{16e}$$

$$\Delta_f > 0 \quad \forall f \tag{16f}$$

$$h_2 \tan\frac{\theta}{2} \le e \le 2h_2 \tan\frac{\theta}{2}. \tag{16g}$$

Some explanations are in order. Constraint (16b) is the constraint of the cache refreshing frequency threshold M. Constraint (16c) is the cache capacity constraint of the edge layer AAVs. Constraint (16d) ensures that the service rate would meet VUs' needs, while (16e) and (16f) are the binary content placement constraint and the nonnegative cache refreshing cycle constraint, respectively. In order to ensure an

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area is served by one (minimum) or two (maximum) adjacent AAVs, the distance between two AAVs at the 2nd layer should satisfy $r_2 \le e \le 2r_2$. According to (1), (16g) is obtained.

The optimization problem **P1** is nonconvex, we utilize the analogy between the CUVIN and the single-BS multinodes multiusers edge caching system and borrow the alternative method introduced in [11] to find the optimal AoI-energy edge caching strategy, while interested readers may refer to [11] for more details. Therefore, by fixing one of the content placement collection $\mathcal{C}$ and the cache refreshing cycle collection Δ_{ref}, **P1** can be decoupled into subproblem **P1.1** and subproblem **P1.2**, which can be solved effectively.

A. Optimization of Cache Refreshing Cycle

Given the content placement collection $\mathcal{C}$, **P1** can be simplified as a cache refreshing cycle optimization subproblem **P1.1** shown as follows:

$$\mathbf{P1.1}: \min_{\Delta_{\text{ref}}} \left\{ \sum_{f=1}^{F} \left(x_1 \Delta_f + y_1/\Delta_f\right) + z_1 \right\} \tag{17a}$$

$$\text{s.t. (16b), (16f) and (16g)} \tag{17b}$$

where $x_1 = \alpha p_f c_f/2$, $y_1 = (1-\alpha)s_f \varepsilon_f c_f/F$, and z_1 admits

$$z_1 = \sum_{f=1}^{F} \frac{\alpha p_f c_f}{\frac{\mathcal{R}_2}{s} - \sum_{f=1}^{F} \Lambda_2^f c_f} + \sum_{f=1}^{F} \alpha p_f \left(1-c_f\right) \Delta_f^{l_1} + \sum_{f=1}^{F} \frac{(1-\alpha) s_f \varepsilon_f \Lambda_1^f \left(1-c_f\right)}{F}. \tag{18}$$

It can be seen that **P1.1** is a convex problem that can be solved using a Lagrange multiplier with Karush-Kuhn-Tucker (KKT) conditions [34].

Lemma 2: Let δ represents the Lagrange multiplier of (16b), the optimal solution of the subproblem **P1.1** satisfies one of (19a) and (19b) as follows

$$\begin{cases} \Delta_f = \sqrt{\dfrac{y_1}{x_1}}; \text{ if } \delta = 0 \text{ and } \displaystyle\sum_{f=1}^{F} \frac{c_f}{\Delta_f} < M & (19a) \\ \sqrt{\dfrac{y_1 + \delta c_f}{x_1}}; \text{ otherwise.} & (19b) \end{cases}$$

Proof: Please refer to Appendix B. ∎

Let $\Delta_f = \sqrt{(\delta c_f + y_1)/x_1}$, if $\delta = 0$ and $\sum_{f=1}^{F} c_f/\Delta_f < M$, the optimal solution is $\Delta_f = \sqrt{y_1/x_1}$. Otherwise, the optimal δ in (19b) has to be solved by dichotomy. In summary, the cache refreshing algorithm in this section can be presented in Algorithm 1. Note that the complexity of Algorithm 1 is $\mathcal{O}(\log_2((u-d)/\varepsilon))$.

Algorithm 1: Cache Refreshing Algorithm

Input: content placement variable c_f, content popularity variable p_f, weighting factor α, unit AAV energy consumption ε_f, cache refreshing frequency threshold M, the dichotomy lower bound d, the dichotomy upper bound u and the dichotomous precision ε.

```
Set Δ_f = sqrt((δc_f + y_1)/x_1);
if δ = 0 and Σ_{f=1}^{F} c_f/Δ_f < M then
    Δ_f = sqrt(y_1/x_1);
else
    Set f(δ) = δ · ( Σ_{f=1}^{F} (c_f/Δ_f) − M );
    p = 1;
    ϖ = (u + d)/2;
    while f(u) · f(d) < 0 and f(ϖ) > ε do
        ϖ = (u + d)/2;
        if f(ϖ) · f(d) < 0 then
            u = ϖ;
        else
            p = p + 1;
            d = ϖ;
        end
    end
end
end
```

Output: Lagrange multiplier δ, cache refreshing cycle collection Δ_{ref}.

B. Optimization of Content Placement

Given the cache refreshing cycle collection Δ_{ref}, **P1** can be simplified as **P1.2** as follows:

$$\mathbf{P1.2}: \min_{\mathcal{C}} \left\{ \frac{\alpha \frac{\mathcal{R}_2}{s}}{\Lambda_2 \left(\frac{\mathcal{R}_2}{s} - \sum_{f=1}^{F} c_f \Lambda_2^f \right)} + \sum_{f=1}^{F} x_2 c_f + y_2 \right\}$$

$$\text{s.t. (16b)-(16e) and (16g)} \tag{20}$$

where

$$x_2 = \alpha p_f \left(\Delta_f/2 - \Delta_f^{l_1}\right) + (1-\alpha) s_f \varepsilon_f \left(1/\Delta_f - \Lambda_1^f\right)/F \tag{21}$$

and $y_2 = \sum_{f=1}^{F} (\alpha p_f \Delta_f^{l_1} + (1-\alpha)\Lambda_1^f s_f \varepsilon_f/F) - (\alpha/\Lambda_2)$.

Due to the binary variable c_f and the nonlinear term $\alpha(\mathcal{R}_2/s)/(\Lambda_2((\mathcal{R}_2/s) - \sum_{f=1}^{F} c_f \Lambda_2^f))$ in (20), **P1.2** is a nonlinear integer programming problem. In order to solve the nonconvexity of the binary variable $c_f \in \{0, 1\}$, it is relaxed into an

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equivalent continuous form $\tilde{c}_f$, which can be shown as

$$\begin{cases} 0 \le \tilde{c}_f \le 1 \quad \forall f & \text{(22a)} \\ \sum_{f=1}^{F} \tilde{c}_f - \sum_{f=1}^{F} \left(\tilde{c}_f\right)^2 \le 0. & \text{(22b)} \end{cases}$$

Denote $\tilde{\mathcal{C}}$ as the collection of continuous relaxation variable $\tilde{c}_f$. By introducing positive penalty factor ψ and nonnegative relaxation variable υ, **P1.2** can be converted to an equivalent subproblem **P1.3** as follows:

$$\mathbf{P1.3}: \min_{\tilde{\mathcal{C}},\upsilon} \left\{ \frac{\alpha \frac{\mathcal{R}_2}{s}}{\Lambda_2\left(\frac{\mathcal{R}_2}{s} - \sum_{f=1}^{F} \tilde{c}_f \Lambda_2^f\right)} + \sum_{f=1}^{F} x_2 \tilde{c}_f + \psi \upsilon \right\} \quad \text{(23a)}$$

$$\text{s.t.} \quad \sum_{f=1}^{F} \tilde{c}_f - \sum_{f=1}^{F} \left(\tilde{c}_f\right)^2 \le \upsilon \quad \text{(23b)}$$

$$\text{(16b)-(16e), (16g), and (22a).} \quad \text{(23c)}$$

The successive convex approximation (SCA) [35] method is utilized, thus the nonconvex term $-\sum_{f=1}^{F}(\tilde{c}_f)^2$ in (23b) is replaced by its first-order Taylor expansion in the χth iteration as

$$-\textstyle\sum_{f=1}^{F} \left(\tilde{c}_f\right)^2 \approx \sum_{f=1}^{F} \left(\tilde{c}_f^{(\chi)}\right)^2 - 2\sum_{f=1}^{F} \tilde{c}_f^{(\chi)} \tilde{c}_f \quad (24)$$

where $\tilde{c}_f^{(\chi)}$ is the value of $\tilde{c}_f$ obtained from the χth iteration. Substituting (24) into (23b), the subproblem **P1.3** can be converted to **P1.4** as follows:

$$\mathbf{P1.4}: \min_{\tilde{\mathcal{C}},\upsilon} \left\{ \frac{\alpha \frac{\mathcal{R}_2}{s}}{\Lambda_2\left(\frac{\mathcal{R}_2}{s} - \sum_{f=1}^{F} \tilde{c}_f \Lambda_2^f\right)} + \sum_{f=1}^{F} x_2 \tilde{c}_f + \psi \upsilon \right\} \quad \text{(25a)}$$

$$\text{s.t.} \quad \sum_{f=1}^{F} \left(1 - 2\tilde{c}_f^{(\chi)}\right) \tilde{c}_f + \sum_{f=1}^{F} \left(\tilde{c}_f^{(\chi)}\right)^2 \le \upsilon \quad \text{(25b)}$$

$$\text{(16b)-(16e), (16g), and (22a).} \quad \text{(25c)}$$

In summary, the content placement algorithm based on SCA can be summarized as Algorithm 2. Note that the complexity of Algorithm 2 is $\mathcal{O}(I \cdot (2)^3)$, where I is the number of iterations of Algorithm 2. Utilizing the MATLAB Convex (CVX) toolbox, **P1.3** and **P1.4** can be solved when Algorithm 3 converges. Due to the core idea of SCA, $\tilde{c}_f$ is approximated in each iteration and the objective value approximates to the upper bound of **P1.3** after each SCA iteration. Finally, Algorithm 3 will converge due to (16b)–(16e).

The AOCS is shown in Algorithm 3. Accordingly, the complexity of Algorithm 3 is $\mathcal{O}(L \cdot (\log_2((u-d)/\varepsilon)) + (I \cdot (2)^3))$, where L is the number of iterations of Algorithm 3.

Algorithm 2: Content Placement Algorithm

Input: content placement variable c_f, content popularity variable p_f, weighting factor α, unit AAV energy consumption ε_f, cache refreshing frequency threshold M and cache refreshing cycle variable Δ_f.

Initialize $\tilde{c}_f^{(0)} = \tilde{c}_f$ and penalty factor ψ;
while *the internal loop stop condition does not meet* **do**
  Solve **P1.4** using CVX to obtain $\tilde{c}_f^{(i)}$;
  set $i = i + 1$;
end

Output: content placement variable $\tilde{c}_f^{(i)}$.

Algorithm 3: Caching Strategy Based on Alternating Optimization

Input: content popularity variable p_f, weighting factor α, unit AAV energy consumption ε_f and cache refreshing frequency threshold M.

Initialize $\tilde{c}_f^{(0)} = 1$;
while *the stop condition of the outer loop does not meet* **do**
  Use **Algorithm 1** to solve the cache refreshing cycle variable Δ_f;
  Initialize $\tilde{c}_f^{(0)} = \tilde{c}_f$ and penalty factor $\psi = 0$;
  Use **Algorithm 2** to obtain the content placement variable $\tilde{c}_f^{(i)}$;
end
return

Output: content placement collection $\mathcal{C}$, cache refreshing cycle collection Δ_{ref}.

V. Simulations and Discussions

In this section, simulation results are presented to validate the effectiveness of the proposed AOCS strategy. To demonstrate the advantage of the proposed AOCS, three existing strategies are compared with AOCS, mainly analyzing the metrics of AoI and energy consumption under different cache refreshing frequency threshold M, density of VUs ρ_v, Zipf distribution coefficient μ of content popularity, total number of content F and AAV cache capacity $S_c^{l_2}$. Simulation parameters are shown in Table I.

Three existing strategies are listed as follows.

1) *Greedy Caching Strategy (GCS) [36]:* In this scheme, the cache refreshing period $\Delta_f = \sqrt{y_1/x_1}$ is fixed.
2) *Uniform Caching Strategy (UCS) [37]:* In this scheme, the caching probability is the same for each content.
3) *Most Popular Content Caching Strategy (MPCS) [38]:* In this scheme, the probability of content caching is equal to the popularity of content.

Figs. 3–7 depict the comparisons among the four different strategies in terms of the weighted sum Ω of the content AoI and energy consumption of the macro AAV u^{l_1}, the AoI Δ of content and the AAV energy consumption Φ. Specifically, in Fig. 3(a), when the threshold of cache refreshing frequency $M = 40$ Hz, the weighted sum Ω of UCS, MPCS, and GCS are about 0.24, 0.30, and 0.31, respectively, while the proposed AOCS strategy reduces Ω to about 0.18. It is worth noting in

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TABLE I
SIMULATION PARAMETERS

Description	Parameter	Values
Number of lanes	J	3
UAV coverage radius [18]	r_i	{300 m, 50 m}
Total number of content	F	120
Density of VUs	ρ_v	0.15 veh/m
UAV cache capability [18]	$S_c^{l_i}$	{300 MB, 120 MB}
Mean value of content size	s	2 MB
The transmit power of the UAV	P_i	{40 dBm, 30 dBm}
Bandwidth of the ith layer UAV-Vehicle links	B_i	{100 Hz, 20 Hz}
Background noise power	σ^2	17 dBm
Attenuation factors of LoS and NLoS links	$\{\eta_{LoS}, \eta_{NLoS}\}$	{1 dB, 10 dB}
Specific environmental factors	$\{a_1, a_2\}$	{4.88, 0.43}
Weighting factor	α	0.5
Energy consumption setting	ε_f	0.1 unit/MB

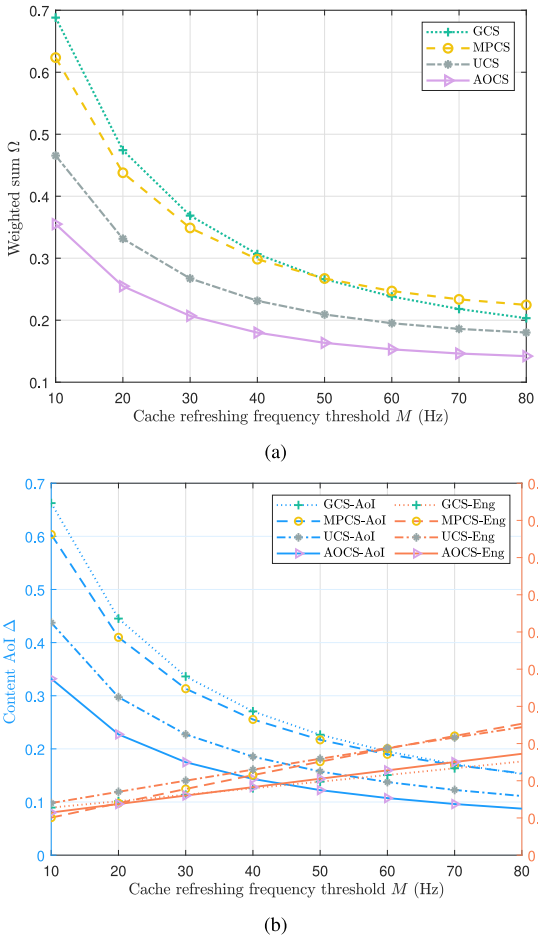


Fig. 3. Different performance indicators versus cache refreshing frequency threshold M. (a) Weighted sum Ω. (b) Content AoI Δ and AAV energy consumption Φ.

Fig. 3(b) that the value of Φ only varies between 0.02 and 0.08 while the value of Δ varies between 0.08 and 0.70. Although the energy consumption of the macro AAV u^{l_1} increases in a linear manner with the increase of M, the increase of the energy consumption is wispy relative to the decrease of the content AoI. Therefore, the weighted sum Ω of Δ and Φ is still on a downward trend. In addition, it can also be observed from Fig. 3(b) that the AoI disparity between the four caching strategies gradually falls off as the increase of M. The reason is that when M is large enough (e.g., at 40 Hz), the content AoI Δ is more related to queuing time rather than caching refreshing time.

The effects of the density of VUs ρ_v are shown in Fig. 4. It can be seen in Fig. 4(a) that under different density of VUs ρ_v, the proposed AOCS strategy substantially outperforms the other three strategies. When $\rho_v = 0.2$ veh/m, the weighted sum Ω of UCS, MPCS, and GCS strategies are about 0.24, 0.30, and 0.31, respectively, while that of the proposed AOCS strategy is only about 0.19. In addition, according to Fig. 4(b), the content AoI Δ and the energy consumption Φ of the macro AAV of the four strategies gradually ascends with the increase of ρ_v. It is worth noting that at a low density of VUs ρ_v, the content AoI Δ depends primarily on the caching refreshing time rather than the queuing time. Under heavy traffic load, the cache is refreshed almost periodically with no change in the refresh interval. In this case, Δ depends mainly on ρ_v, which reflects the request arrival rate and the queue time.

According to [39], content requests follow the Zipf distribution, and the parameter of Zipf distribution may vary between 0.65 and 0.85 depending on the network framework. For a large Zipf distribution coefficient, the popular content will likely account for a large percentage of total requests. According to Fig. 5(a), the Ω declines with the growth of the Zipf distribution parameter, that is content popularity coefficient μ. It is shown that the proposed AOCS outperforms more substantially than other schemes when μ is small, which means that AOCS reduces more AoI when content popularity is relatively uniform. When $\mu = 0.55$, the weighted sum of UCS, MPCS, and GCS strategies are about 0.27, 0.32, and 0.39, respectively, while that of the proposed AOCS strategy is only about 0.23. In addition, the AOCS can also achieve the goal that the edge caching layer AAVs cache the most popular content with low Ω to handle the high-arrival rate caused by frequent requests for popular content. Furthermore, since the change in AAV energy consumption Φ is small relative to the change of content AoI Δ, to analyze the influence of μ on Φ, the changes in AAV energy consumption under different μ are plotted in Fig. 5(b). When $\mu = 1.15$, the value of AAV energy consumption Φ of UCS, MPCS, and GCS strategies are about 0.046, 0.044, and 0.037, respectively, while that of the proposed AOCS strategy is only about 0.036. Although the energy consumption of the AOCS is not always the lowest, it offers the best balance between AoI and energy consumption.

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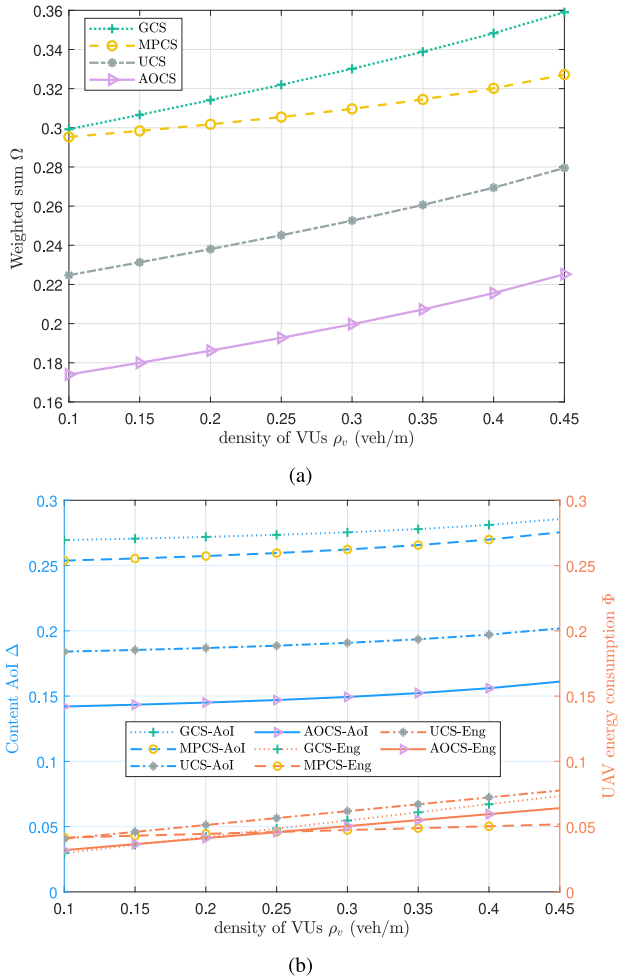


Fig. 4. Different performance indicators versus density of VUs ρ_v. (a) Weighted sum Ω. (b) Content AoI Δ and AAV energy consumption Φ.

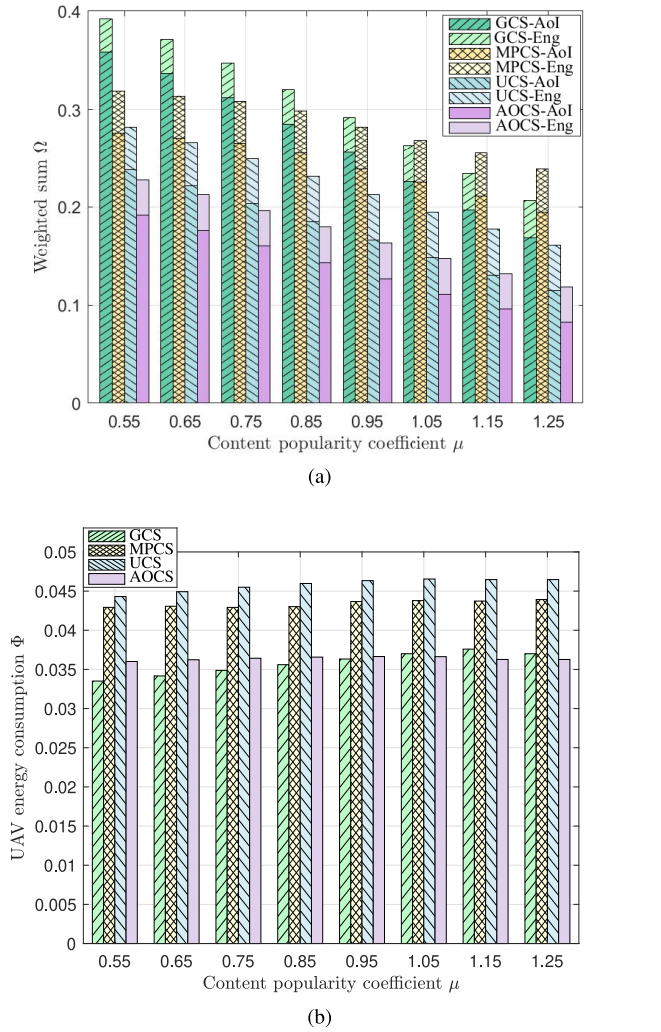


Fig. 5. Different performance indicators versus content popularity coefficient μ. (a) Weighted sum Ω. (b) AAV energy consumption Φ.

The performance of these strategies under different AAV cache capacity $S_c^{l_2}$ is compared in Fig. 6. As seen from Fig. 6(a), when $S_c^{l_2} = 90$ MB, the weighted sum Ω of UCS, MPCS, and GCS strategies are about 0.17, 0.23, and 0.25, respectively, while that of the proposed AOCS strategy is only about 0.12. The Ω increases when $S_c^{l_2}$ grows, which means the network performance declines. It can be observed from Fig. 6(b) that the content AoI Δ gradually ascends with the increase of AAV cache capacity $S_c^{l_2}$. The reason is that the total amount of content cached in the edge layer AAVs rises as $S_c^{l_2}$ grows, i.e., longer content caching and delivery time, resulting in higher AoI. Indirectly, the total amount of content fetched directly from the macro AAV u^{l_1} decreases, so the part of energy consumption generated by delivering content falls while the part of energy consumption for u^{l_1} to refresh the cache of the edge layer AAVs goes up. As a result, the total energy consumption Φ of the macro AAV u^{l_1} slightly fluctuates between 0.03 and 0.05 due to the increase and decrease in energy consumption partially offset each other. Similarly, the proposed AOCS strategy still results in better performance compared to the other three strategies.

Finally, the weighted sum Ω and the changing trend of four strategies under the different total number of content F are shown in Fig. 7(a), while Fig. 7(b) and (c) depict the AoI Δ of content and energy consumption Φ of the macro AAV u^{l_1} under different $S_c^{l_2}$ and F, respectively. It can be seen from Fig. 7(a) and (b) that the performance gain brought by the proposed AOCS strategy is significant. For example, in Fig. 7(a), when $S_c^{l_2} = 60$ MB and $F = 60$, the weighted sum Ω of UCS, MPCS, and GCS strategy are about 0.23, 0.24, and 0.25, respectively, while that of the proposed AOCS strategy is only about 0.21. What can be seen from Fig. 7(c) is that when F ascends, Φ shows a significant downward trend, but its change in terms of $S_c^{l_2}$ is slight, which also confirms the conclusion of Fig. 6(b). In particular, the four strategies behave consistently when $F \leq 40$ and $S_c^{l_2} = 100$ MB. The reason for this phenomenon is that the edge layer AAVs can cache all the content in this case, the four strategies have the same content placement collection and caching refreshing cycle collection. Similarly, the four strategies behave uniformly when $F \leq 60$ and $S_c^{l_2} = 140$ MB.

As shown in Fig. 8, the weighted sum Φ increases with the increase of the weighting factor α, and the superiority of the proposed AOCS is also verified. This is because the AOCS optimizes the cache refreshing and the content placement

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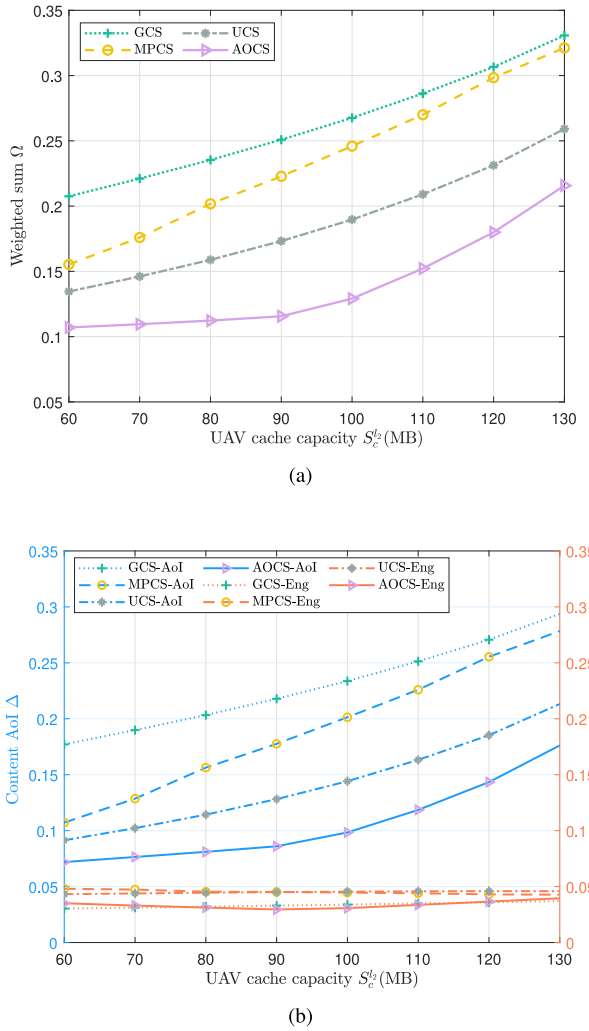


Fig. 6. Different performance indicators versus capacity $S_c^{l_2}$ of the edge layer AAV. (a) Weighted sum Ω. (b) Content AoI Δ and AAV energy consumption Φ.

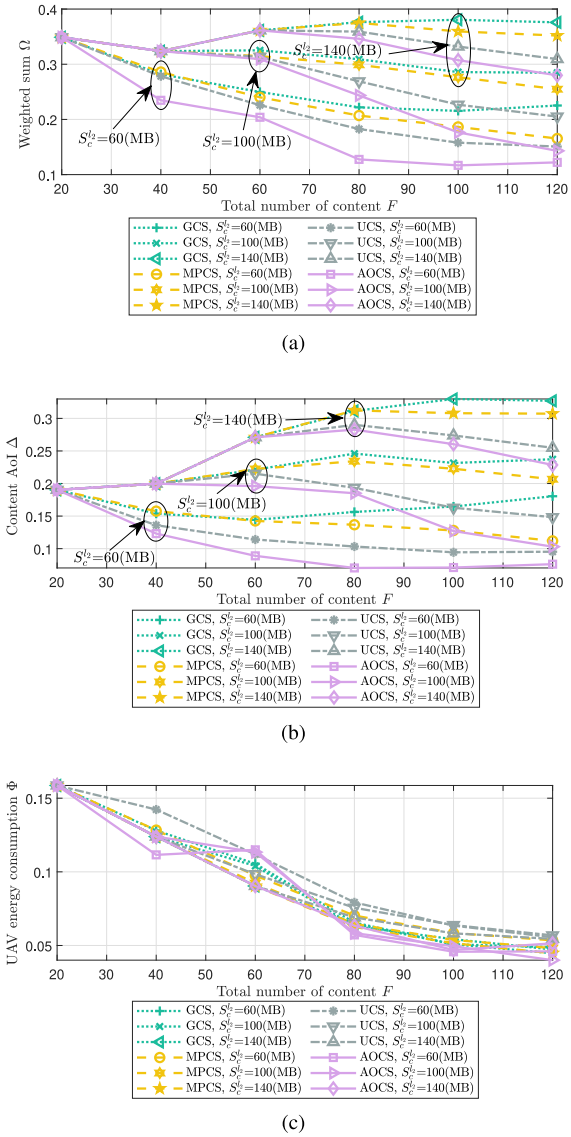


Fig. 7. Different performance indicators versus total number of content F. (a) Weighted sum Ω. (b) Content AoI Δ. (c) AAV energy consumption Φ.

jointly, while the others either ignore the cache refreshing optimization or focus on unitary content placement.

VI. Conclusion and Future Works

In this article, a joint optimization problem in the CUVIN has been proposed, in which a caching strategy based on alternating optimization has been designed to provide the minimum weighted sum of the content AoI and the macro AAV's energy consumption. Due to the coupling variables and the binary nonconvex constraint of the proposed optimization problem, it has been decoupled into two subproblems. Specifically, for the content placement subproblem, the locally optimal solution has been obtained through SCA. For the cache refreshing cycle subproblem, the optimal solution of the semi-closed form has been derived with the KKT conditions. Simulation results have shown that the proposed AOCS strategy can significantly improve the freshness of content and reduce the energy consumption of the macro AAV in the CUVIN.

Worthwhile future works include studying the total energy consumption of both the edge layer AAVs and the macro AAV of the proposed strategy, incorporating mobility of users and AAVs, allowing mobile edge computing at the same time, and integrating stochastic geometry to derive the success probability.

Appendix A
Proof of Lemma 1

Proof: For a given number N_2 of VUs' requests, the average conditional difference time for content f_f is expressed as [11]

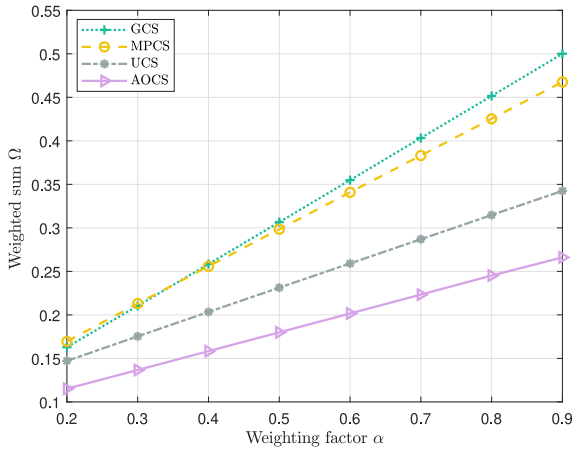
$$\mathbb{E}\left(\Delta_f^d\Big|_{\mathcal{N}=N_2}\right)=\mathbb{E}\left(\frac{1}{N}\sum_{n=1}^{N_2}t_n\mid\mathcal{N}=N_2\right)=\frac{\Delta_f}{2} \quad (26)$$

where $\mathcal{N}$ is the number of requests for content f_f from VUs during the cache refreshing cycle Δ_f.

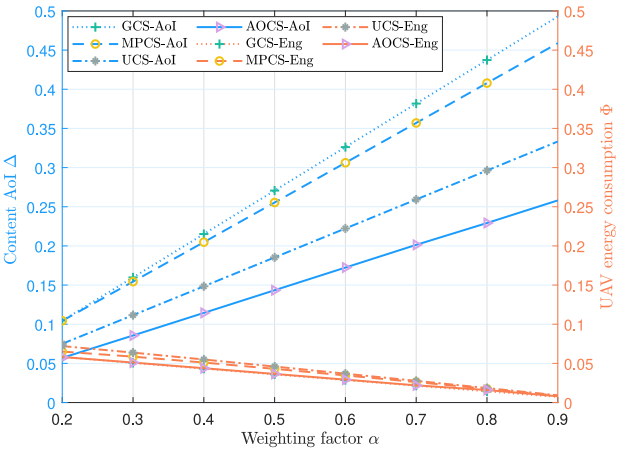
Considering all possible VUs' requests, the average difference time for content f_f can be written as follows:

$$\mathbb{E}\left[\Delta_f^d\right]=\sum_{N_2=0}^{\infty}\left[P(\mathcal{N}=N_2)\mathbb{E}\left(\Delta_f^d\Big|_{N=N_2}\right)\right]=\frac{\Delta_f}{2} \quad (27)$$

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(a)



(b)

Fig. 8. Different performance indicators versus weighting factor α. (a) Weighted sum Ω. (b) Content AoI Δ and AAV energy consumption Φ.

this is because $\sum_{N_2=0}^{\infty} P(\mathcal{N}=N_2)=1$.

The average service latency of content served by the second layer AAV u_k^{l2} is expressed as

$$\mathbb{E}\left[\Delta_{2,wf}\right]=\frac{1}{\frac{\mathcal{R}_2}{s}-\sum_{f=1}^{F}\Lambda_2^f\cdot c_f}=\frac{1}{\frac{\mathcal{R}_2}{s}-\sum_{f=1}^{F}2Jr_2\rho_v\lambda p_f c_f}. \tag{28}$$

Substituting (26) and (27) into (11), Lemma 1 is proved. ■

APPENDIX B
PROOF OF LEMMA 2

Proof: The Lagrangian function (see [11, Proposition 1]) of the subproblem **P1.1** is expressed as

$$\mathcal{L}(\Delta_f,\delta)=\sum_{f=1}^{F}\left(x_1\Delta_f+\frac{y_1}{\Delta_f}\right)+\delta\left(\sum_{f=1}^{F}\frac{c_f}{\Delta_f}-M\right)+z_1 \tag{29}$$

the derivative of $\mathcal{L}(\Delta_f,\delta)$ with respect to Δ_f is given by

$$\frac{\partial\mathcal{L}(\Delta_f,\delta)}{\partial\Delta_f}=x_1-\frac{y_1}{(\Delta_f)^2}-\frac{\delta c_f}{(\Delta_f)^2}\quad\forall f. \tag{30}$$

Then, $\partial\mathcal{L}(\Delta_f,\delta)/\partial\Delta_f=0$ and $\Delta_f>0$ ought to be satisfied. It can be concluded as

$$\Delta_f=\sqrt{(\delta c_f+y_1)/x_1}\quad\forall f. \tag{31}$$

Accordingly, the KKT conditions for the subproblem **P1.1** are presented as

$$\begin{cases}\Delta_f=\sqrt{(\delta c_f+y_1)/x_1}\quad\forall f\\ \sum_{f=1}^{F}\frac{c_f}{\Delta_f}<M\\ \delta\geq 0\\ \delta\left(\sum_{f=1}^{F}\frac{c_f}{\Delta_f}-M\right)=0.\end{cases} \tag{32}$$

In particular, there are two cases with respect to δ as follows.

1) When $\delta=0$, $\Delta_f=\sqrt{y_1/x_1}$, if $\sum_{f=1}^{F}c_f/\Delta_f<M$ holds, the optimal solution of **P1.1** is

$$\Delta_f=\sqrt{y_1/x_1},\delta=0. \tag{33}$$

2) When $\delta>0$, the optimal solution of **P1.1** is $\Delta_f=\sqrt{(\delta c_f+y_1)/x_1}$. In this case, complementary relaxation conditions need to be met

$$\sum_{f=1}^{F}(c_f/\Delta_f)-M=0. \tag{34}$$

Hence, Lemma 2 is proved. ■

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